Math 101 — Final Exam Extra Credit

Part 1 (1 point): Complete all 20 provided questions on separate paper. Show handwritten steps. Red answers are for checking only.

Part 2 (1 point): Create and solve 30 similar questions, showing handwritten work. You may ask for help but must submit your own work.

Total extra credit: 2 points. Final extra credit is limited to 2 points because of department guidelines.

1. Find the vertex and axis of $y = x^2 - 4x + 1$.

Answer: $(2, -3)$; $x = 2$

2. For $f(x) = \sqrt{x+8}$ and $g(x) = x - 2$, find $f(g(x))$.

Answer: $\sqrt{x+6}$

3. Evaluate $\log_2\left(\frac{1}{8}\right)$.

Answer: -3

4. Find a line perpendicular to $y = 2x + 3$ through $(2, 1)$.

Answer: $y = -\frac{1}{2}x + 2$

5. For $y = -2^x + 1$, state the horizontal asymptote and y-intercept.

Answer: Asymptote $y = 1$; $(0, 0)$

6. Solve $2x + 1 < -3$ OR $3x - 2 \geq 7$. Give interval notation.

Answer: $(-\infty, -2) \cup [3, \infty)$

7. Simplify $\frac{x^2-9}{x^2+5x+6}$. State excluded values.

Answer: $\frac{x-3}{x+2}$; $x \neq -3, -2$

8. Divide $\frac{4+3i}{1-i}$. Give $a + bi$ form.

Answer: $\frac{1}{2} + \frac{7}{2}i$

9. For $f(x) = \begin{cases} x+1, & x < 2 \\ 4, & x \geq 2 \end{cases}$, state $f(2)$ and the open endpoint.

Answer: $f(2) = 4$; open at $(2, 3)$

10. Rationalize $\frac{5}{\sqrt{3}}$.

Answer: $\frac{5\sqrt{3}}{3}$

11. Simplify $\left(\frac{x^{-2}y^3z^{-1}}{x^2y^{-1}z^2}\right)^{-1}$.

Answer: $\frac{x^4z^3}{y^4}$

12. Solve $2x^2 - 3x - 2 = 0$.

Answer: $x = 2, -\frac{1}{2}$

13. Solve $\frac{1}{x-1} = \frac{2}{x+2}$.

Answer: $x = 4$; excluded $x = 1, -2$

14. Multiply and simplify $\frac{x^2-4}{x+3} \cdot \frac{x+3}{x-2}$.

Answer: $x + 2$; excluded $x = -3, 2$

15. Solve $2x + 3 = 11$.

Answer: $x = 4$

16. A rocket follows $h(t) = -5t^2 + 30t + 20$. Find its maximum height and time.

Answer: 65 units at $t = 3$

17. A cleaner takes 6 hours alone. Together with another cleaner, they take 2 hours. Find the other cleaner's time.

Answer: 3 hours

18. Gas volume varies inversely with pressure. $V = 3000\,\text{cm}^3$ at $P = 400\,\text{mmHg}$. Find V at $600\,\text{mmHg}$.

Answer: $2000\,\text{cm}^3$

19. A 96-inch tube is cut into pieces, one 3 times the other. Find both lengths.

Answer: 24 inches and 72 inches

20. Mix 20% and 50% solutions to make 100 mL of 32% solution. How much of each?

Answer: 60 mL of 20%; 40 mL of 50%